Conformal Prediction

ELG 5218 - Uncertainty Evaluation in Engineering Measurements and Machine Learning

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Outline: Conformal Prediction

- **Motivation & Goal** (distribution-free uncertainty sets)
- **Split conformal: the basic algorithm** (classification & regression)
- **Finite-sample coverage guarantee** (why it works)
- **Efficiency** (set size / interval length) + practical choices
- **Applications**:
  - Image classification (prediction sets)
  - Object detection (bounding-box regions)
  - LLM-style QA (sets + abstention)
- **Notebook & assignment**

Motivation: What do we want?

- A trained model gives a point prediction $\hat{y}(x)$ (or a softmax distribution).
- We often need **valid uncertainty**:
  - Regression: prediction interval that contains y with probability $\geq 1 - \alpha$.
  - Classification: prediction **set** that contains the true label with probability $\geq 1 - \alpha$.
- **Key challenge**: we want guarantees with minimal assumptions.
- **Conformal prediction**: a *wrapper* around any model that gives **finite-sample marginal coverage** under i.i.d./exchangeability.

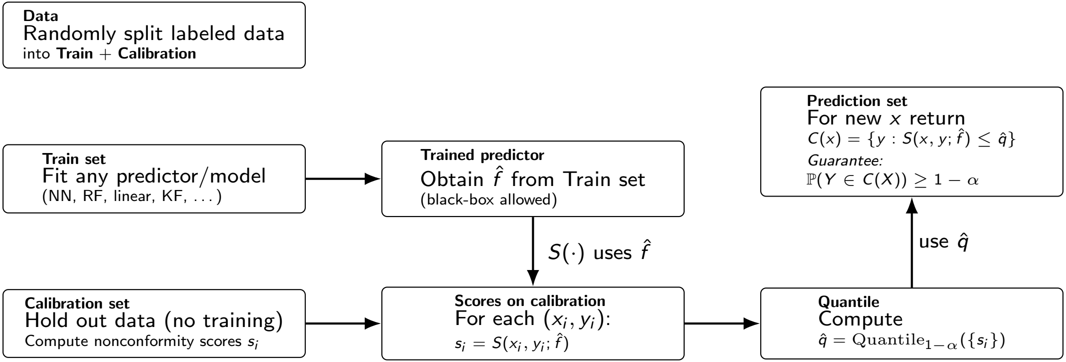
One-line summary

Split conformal recipe

Train any predictor on a training set. On a held-out *calibration* set, compute nonconformity scores s_i. Let $\hat{q}$ be the $(1-\alpha)$ quantile (order statistic). For a new x, return the set/interval of all y whose score is $\leq \hat{q}$.

- Coverage is **guaranteed** (marginally) regardless of model correctness.
- Efficiency depends on model quality + score design.

Split conformal in one picture



Split conformal: algorithm (classification)

Inputs: trained model producing probabilities $p(y \mid x)$, calibration set $\{(x_i, y_i)\}_{i=1}^{n}$.

Choose a nonconformity score $s(x, y)$ (bigger = "less conforming"). Common examples:

- **Threshold score**: $s(x, y) = 1 - p(y \mid x)$
- **APS score**: $s(x, y) = \sum_{y' : p(y' \mid x) \geq p(y \mid x)} p(y' \mid x)$ (often smaller sets)

Calibrate: compute $s_i = s(x_i, y_i)$, set $\hat{q} = \mathrm{Quantile}_{1-\alpha}(\{s_i\})$.
Predict: return $T(x) = \{y \in \mathcal{Y} : s(x, y) \leq \hat{q}\}$.

Split conformal: algorithm (regression)

Model: $\hat{y}(x)$.
Score: $s(x, y) = |y - \hat{y}(x)|$ (absolute residual).
Calibrate: $s_i = |y_i - \hat{y}(x_i)|$ on calibration set, $\hat{q} = \mathrm{Quantile}_{1-\alpha}(\{s_i\})$.
Predict interval:

$$C(x) = [\hat{y}(x) - \hat{q},\ \hat{y}(x) + \hat{q}].$$

Interpretation: $\hat{q}$ is the smallest half-width that yields the target coverage on calibration (with finite-sample correction).

Finite-sample coverage guarantee (marginal)

Guarantee (informal)

If data are i.i.d. (exchangeable) and we use split conformal with calibration size n, then

$$\Pr\left(Y \in T(X)\right) \geq 1 - \alpha$$

(up to a small finite-sample correction typically handled by the $(n+1)$ order statistic).

- No assumptions on the data distribution or the model class.
- What is guaranteed: **marginal** coverage (averaged over the joint distribution).
- What is *not* guaranteed by default: conditional coverage for each x.

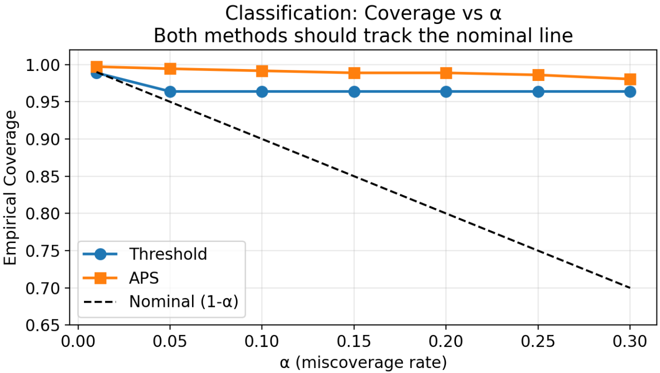
Why it works (intuition)

- Think of the (unknown) test example as being "inserted" into the calibration set.
- Under exchangeability, the rank of the test score among $\{s_1, \ldots, s_n, s_{n+1}\}$ is uniform.
- We choose $\hat{q}$ so that only an α fraction of ranks exceed it.
- Therefore, the probability the test score exceeds $\hat{q}$ is at most α.

Efficiency: prediction set size and interval length

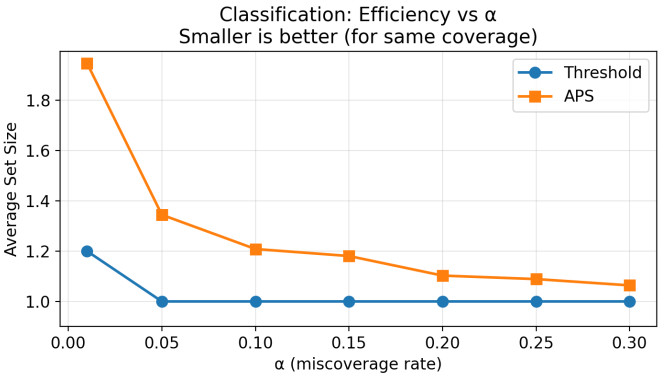
- Coverage alone is not enough: $\mathcal{Y}$ (all labels) gives 100% coverage but is useless.
- Efficiency metrics:
  - Classification: average set size $\mathbb{E}[|T(X)|]$.
  - Regression: average interval length $\mathbb{E}[|C(X)|]$.
- Better models and better scores $\Rightarrow$ smaller sets for the same coverage.
- APS is often more efficient than naive thresholding.

Application 1: Image classification (Digits) — coverage



Slide note: This plot is generated by the course notebook (`conformal_prediction_lab_notebook.ipynb`).

Application 1: Image classification (Digits) — efficiency



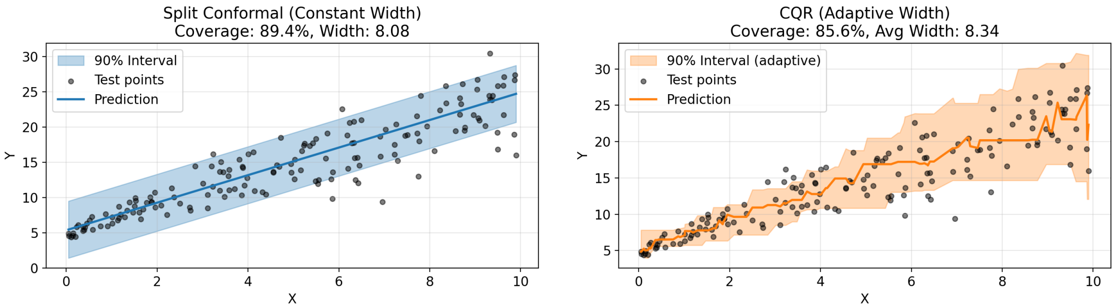
Note: Efficiency must be read alongside coverage. Here, simple thresholding produces smaller sets than APS, but the companion coverage plot suggests it may under-cover for larger α, while APS is conservative (high coverage, larger sets).

ImageNet Example

Figure 1: Prediction set examples on Imagenet. *We show three progressively more difficult examples of the class* ***fox squirrel*** *and the prediction sets (i.e., $\mathcal{C}(X_{\text{test}})$) generated by conformal prediction.*

Source: Anastasios N. Angelopoulos and Stephen Bates (2022)

Regression: conformal prediction intervals

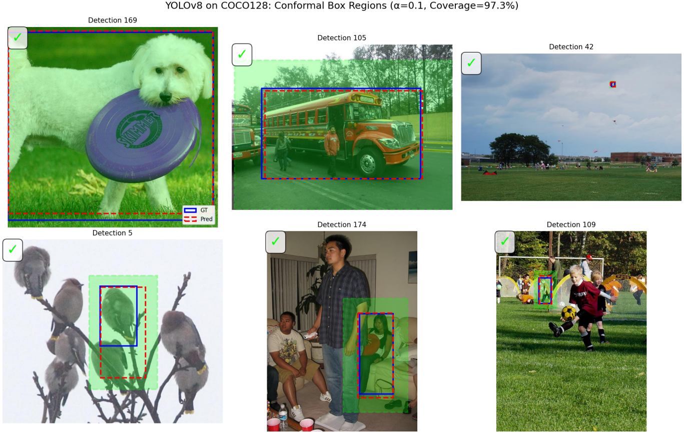


Intervals are constant-width here (residual-based). More advanced variants (e.g., conformalized quantile regression) adapt widths to x.

Application 2: Object detection (bounding-box regions)

- Detection output is structured: class + box + matching.
- A teachable conformal pattern:
  - Match predicted box $\hat{b}$ to ground-truth b on calibration set.
  - Score $s = \|b - \hat{b}\|_\infty$.
  - Conformalize by expanding $\hat{b}$ by $\hat{q}$ in all directions.
- This yields a **box region** that contains the true box with probability $\geq 1 - \alpha$ (marginally).

Object detection: example figures



Green shaded regions = conformal uncertainty regions
Blue = ground truth, Red dashed = predictions

Real Object Detection Example

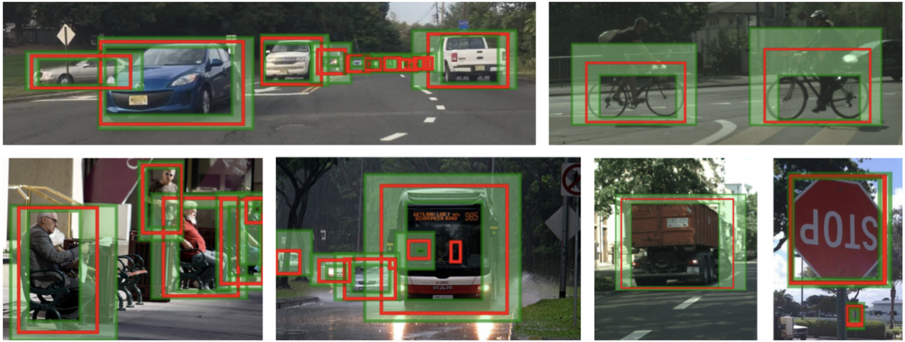


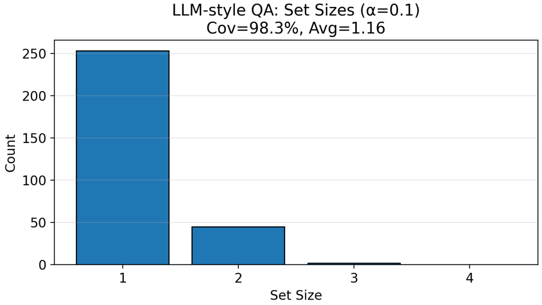
Fig. 1: Examples of our method for multiple classes on test images. True bounding boxes are in red, two-sided prediction interval regions are shaded in green. Produced uncertainty estimates come with a probabilistic coverage guarantee of the true boxes.

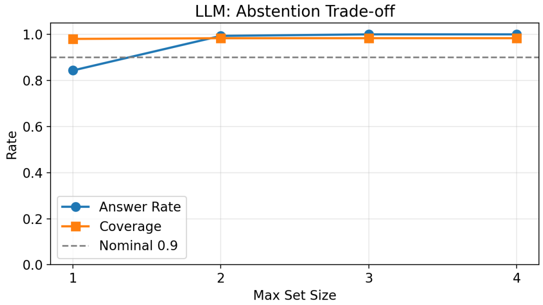
Source: A. Timans et al. (2024)

Application 3: LLM-style QA — prediction sets

- Treat a question as choosing among candidate answers (multiple choice, or ranked generations).
- Use APS on answer probabilities to get a set $T(x)$ with coverage $\geq 1 - \alpha$.
- Then use **abstention**: answer only when $|T(x)|$ is small (e.g., singleton).

LLM-style: set sizes and abstention trade-off





Course notebook (figures used in this deck)

- Notebook: `conformal_prediction_lab_notebook.ipynb`
- Figures exported by the notebook and used here:
  - `digits_conformal_coverage.png`, `digits_conformal_setsize.png`
  - `regression_conformal_intervals.png`
  - `detection_conformal_boxes.png`
  - `llm_conformal_setsizes.png`, `llm_conformal_abstention_tradeoff.png`
- Recommended external repo for richer deep-learning demos: aangelopoulos/conformal-prediction

Assignment

1. **Classification:** compare threshold vs APS sets on Digits at multiple α values.
2. **Regression:** verify empirical coverage vs. α and report average interval length.
3. **Shift test:** calibrate on clean data, test on a shifted version (noise/rotation); analyze what breaks.
4. **Bonus:** implement a simple Mondrian conformal variant (calibrate per class or per difficulty bucket).

Wrap-up

- Conformal prediction gives **distribution-free, finite-sample** marginal coverage.
- The algorithm is simple: score on calibration set + quantile threshold.
- Practical success depends on **score design** and model quality (efficiency).
- Conformal ideas extend naturally to structured outputs and selective prediction (abstention).

Key references (for reading)

- Shafer & Vovk. *A Tutorial on Conformal Prediction*. JMLR 2008.
- Angelopoulos & Bates. *A Gentle Introduction to Conformal Prediction*. (tutorial + code repo).
- Murphy. *Probabilistic Machine Learning: Advanced Topics*. Chapter on conformal prediction.
- Romano et al. *Conformalized Quantile Regression*. NeurIPS 2019.
- Barber et al. *Predictive Inference with the Jackknife+*. NeurIPS 2019.
- Timans et al. *Adaptive Bounding Box Uncertainties via Two-Step Conformal Prediction*. ECCV 2024.
- Kumar et al. *Conformal Prediction with LLMs for Multi-Choice QA*. ICML 2023 workshop.
- A survey: *Conformal Prediction for Natural Language Processing*. TACL 2024.